Lecture 7: Rates & Energy continued

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1 Lecture 7: rates & energy catchup

2 How fast do (eukaryotic cells) degrade proteins

A calculation in the book (Fig. 2) proceeds by estimating the number of proteasomes in a HeLa cell and assuming a typical rate of ≈ 5 proteins degraded / minute / proteasome complex.

The proteasome is a large protein complex (long dimension $\approx f \times 10$ nm, Fig. 1). The proteasome has to be large because it degrades proteins by feeding them through a central pore on the caps (yellow). The proteins are unfolded as they are fed into a large central cavity where the peptide bond hydrolysis (degradation) proceeds. This releases peptides which are broken down to free amino acids that can be reused for new protein synthesis.

The lifecycle of free amino acids (AA) is:

$$\text{AA} \underbrace{\longrightarrow}_{\text{tRNA synthetases}} \text{amino-acyl tRNAs} \underbrace{\longrightarrow}_{\text{ribosome}} \text{proteins} \underbrace{\longrightarrow}_{\text{proteasome}} \text{peptides} \underbrace{\longrightarrow}_{\text{peptidases}} \text{AA}$$

3 Is the proteasome fast or slow?

Each proteasome can degrade ≈ 5 proteins/min. This reaction produces peptides of length ≈ 10 amino acids in length.

How many hydrolysis reactions does the proteasome perform per second?

$$\approx 5 \text{ proteins/min} \times \frac{1\text{min}}{60\text{s}} \times 350 \text{ bonds/protein} \times \frac{1\text{peptide}}{10 \text{ bonds}} \approx 3 \text{ reactions/s}$$

What is the turnover time of proteins through active degradation?

we denote the number of amino acids per cell by N_{aa}

(e.g. for HeLa cell, $N_{aa} \approx 3\text{x}10^6 \frac{\text{proteins}}{\mu\text{m}^3} \times 3000 \frac{\mu\text{m}^3}{\text{cell}} \times 400 \frac{\text{aa}}{\text{protein}} \approx 4\text{x}10^{12} \frac{\text{aa}}{\text{cell}}$)

$\approx 1\%$ of proteome mass is proteasomes

$$\text{number of proteasomes} \approx \frac{0.01 \times N_{aa}}{20{,}000 \text{ aa/proteasome}} \approx 0.5\text{x}10^{-6} N_{aa} \text{ proteasomes/cell}$$

molecular mass of proteasome $\approx$ 2.4 MDa $\approx$ 20,000 aa

proteasome deg. rate $\approx$ 5 proteins/min $\approx$ 0.1 protein/s $\approx$ 40 aa/s

$$\text{total deg. rate} \approx 40 \frac{\text{aa}}{\text{s x proteasome}} \times 0.5\text{x}10^{-6} N_{aa} \frac{\text{proteasomes}}{\text{cell}} \approx 20\text{x}10^{-6} N_{aa} \frac{\text{aa}}{\text{s x cell}}$$

$$\text{turnover time} \approx \frac{\text{number of aa per cell}}{\text{total deg rate}} \approx \frac{N_{aa} \text{ aa}}{N_{aa} \text{ x } 20\text{x}10^{-6} \text{ aa/s}} \approx 0.5\text{x}10^5 \text{s} \approx \boxed{1 \text{ day}}$$

i.e. it would take all the proteasomes working at full speed about one day to degrade all of the proteome.

Figure 2: Estimating the protein turnover in HeLa cells that is due to proteasomes from Cell Biology by The Numbers.

Notice that we used 350 amino acids / protein, which is a number typical for yeast cells, as shown in Figure 3. Bacteria degrade proteins with a related protein complex called ClpXP.

Outline for whiteboard

- HW #3 – take a look
- proteasome
- response time
- jiggling
- binding
- specificity
- projects

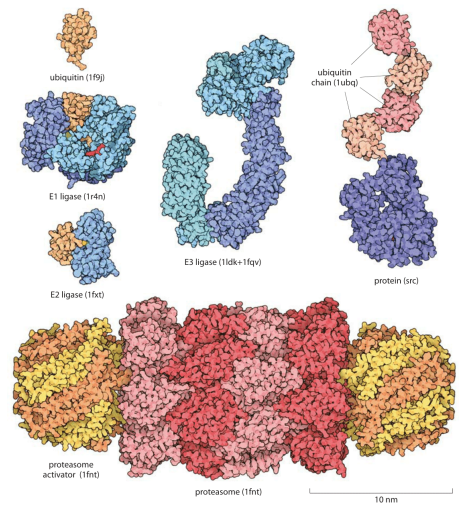

Figure 1: Illustration of the structures of the proteasome and ubiquitination / deubiquitanation machinery from Cell Biol. by the Numbers.

Figure 3: Distribution of protein lengths for different species. From Cell Biology by The Numbers.

3.1 Comparison to typical enzymatic rates

Is 3 reactions/s fast or slow? Looking at Fig 6 it's about threefold lower than the maximum per-active site catalytic rate (k_{cat}) of the median enzyme (≈ 10 /s). Looking at Fig. 7, however, its clear that central metabolic enzymes have higher k_{cat} values around 80 /s.

What would it be like if bacteria grew using enzymes that catalyzed 3 reactions/s?

We previously estimated *E. coli* must consume $f \times 10^6$ glucose/s to support its growth (1 hr generation). Suppose growth is supported by a linear metabolic pathway catalyzed by $N \approx 100$ enzymes operating in serial as in 4. Assume each of the enzymes catalyzes $v = 3$ reactions/s/enzyme.

For the whole pathway to process $f \times 10^6$ glucose/s, each enzyme would need to be present in $f \times 10^6/v \approx 10^6$ copies. If there are 100 such enzymes, then we need 10^7 enzymatic proteins/cell to make this happen.

But we also remember that a cell contains $f \times 10^6$ proteins/fL, about 50% of which are metabolic. So enzymes need to be faster than 3 reactions/s.

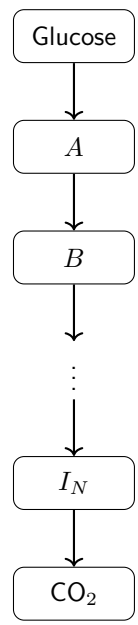

Figure 4: A linear metabolic pathway with N+1 steps

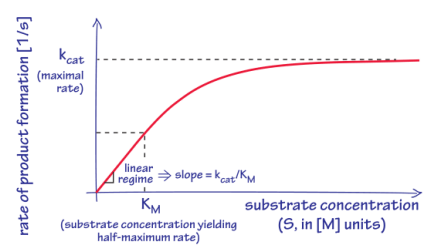

Figure 5: In high substrate concentrations, enzymatic rates saturate at a maximum per-active site rate called the k_{cat}. Graphic from Cell Biology by the Numbers.

4 Typical enzyme rates

How fast are typical enzymes? We have lots of data on enzyme kinetics, which provides an opportunity to comment.

Inspecting Figure 6 we see that a typical enzyms have maximum per active site rates $k_{cat} \approx 10$ reactions/s per active site. This is already **threefold greater than 3 reactions/s** and reduces our estimate of metabolic enzyme burden to $f \times 10^6$ proteins/fL.

Moreover, central metabolic enzymes (the kinds that metabolize glucose) have $k_{cat} \approx 10^2$ reactions/s (see Figure 7). These values produce an estimate of $f \times 10^5$ enzymatic proteins/fL, **flipping the script**.

Now we instead ask: why are metabolic proteins so highly expressed? It seems

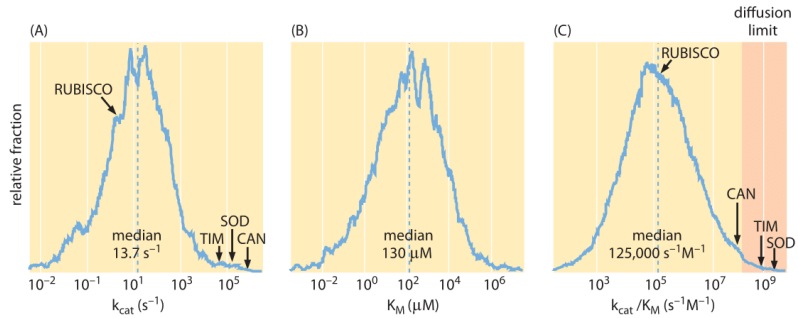

Figure 6: The distribution of enzyme kinetic parameters from the BRENDA database as curated by Bar-Even et al. Biochemistry 2011. Graphic from Cell Biology by the Numbers.

like they only need to be 10% of the proteome, but they are instead $\approx 50\%$.

4.1 Potential biochemistry/physiology projects

These distributions prompt us to consider enzymes that are "too fast" or "too slow" given the evolutionary and physiological pressures. Famous examples of metabolic enzymes that are slower than we might expect are rubisco and nitrogenase. Can you think of examples that are "too fast?"

It's also useful to remember that many regulatory proteins are enzymes. mTor is a regulatory protein kinase that controls metabolism in eukaryotic cells – it catalyzes the phosphorylation of target proteins with ATP as the phospho-donor. Another famous mammalian regulator, Ras, is also an enzyme: it's a GTPase. **Should we expect these regulators to be fast or slow? How fast are they?**

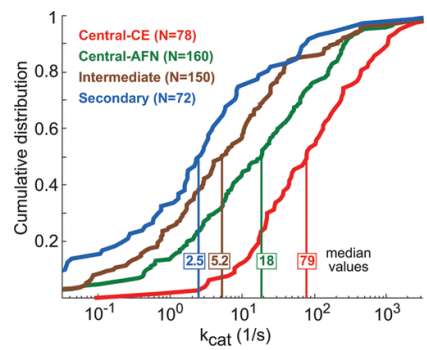

Figure 7: The distribution of k_{cat} values curated by Bar-Even et al. Biochemistry 2011.

5 Biology and catalysis

Life is distinguished from non-life in many ways, but several of them are related to catalysis.

Some comments on the role of catalysis in organisms. These apply across the tree of life and are relevant across scales.

1. Many relevant reactions are intrinsically slow ($\ll 1$ s^{-1}) without catalysis.
2. Life uses catalysts (usually proteins) to speed reactions up.
3. Rate enhancements k_{cat}/k_{uncat} can exceed 10^{10} fold (Wolfenden 2001).
4. Catalysis accelerates metabolism, regulation, and sensing.
5. Regulation of catalysis puts cells/organisms in control of those processes.
6. Reactions that occur spontaneously and rapidly are not controllable.

6 Response time and turnover of proteins

How long does it take for new protein expression to produce a meaningful physiological response? Let's take an example from insulin signaling.

mTORC1[1] is activated by insulin. This causes the expression and nuclear accumulation of SREBP-1c (Sterol Regulatory Element-Binding Protein-1c), which

[1] a protein complex containing mTor

facilitates storage of fatty acids (see Wikipedia). **How long should it take for SREBP-1c to accumulate in the nucleus?**

6.1 Protein Induction Dynamics

Following induction (e.g., insulin addition), the protein concentration obeys:

$$\frac{dc}{dt} = \beta - \alpha\, c, \qquad \alpha = \alpha_{\text{deg}} + \alpha_{\text{dil}}$$

α_{deg} is the rate at which proteins are degraded and α_{dil} is the rate at which they are diluted by growth.

The steady-state level $c_{\text{st}} = \beta/\alpha$ is derived by setting $\frac{dc}{dt} = 0$. For linear homogeneous differential equations like above, the solution has an exponential form when the coefficients are constants, as they are here (α and β).

Here's exponential solution that satisfies our $c(0) = 0$ and $c(\infty) = c_{\text{st}}$:

$$c(t) = c_{\text{st}}\left(1 - e^{-\alpha t}\right).$$

It's good to check the boundary conditions:

$$c(0) = c_{\text{st}} \cdot (1 - 1) = 0 \qquad c(\infty) = c_{\text{st}} \cdot (1 - 0) = c_{\text{st}}.$$

So how long does it take to get to c_{st}? Solve for $t_{1/2}$

$$c(t_{1/2}) = \frac{1}{2} c_{\text{st}} = c_{\text{st}} \cdot \left(1 - e^{-\alpha t_{1/2}}\right) \implies t_{1/2} = \ln(2)/\alpha.$$

Some comments

- Steady-state level: $c_{\text{st}} = \beta/\alpha$
- Response time: $\tau_{\text{response}} \propto 1/\alpha$ ind. of β.
- $\tau_{\text{response}} \approx \min(\tau_{\text{deg}}, \tau_{\text{cell cycle}})$

Key insight: The synthesis rate β does *not* affect the response time — it only sets the final steady-state level. A higher β scales the curve up, but the time constant $1/\alpha$ is unchanged.

Implications: can shorten response time by spending energy on proteasomal degradation of a protein. This leads us to suspect regulated degradation of SREBP-1c.

More broadly: this logic applies in many places in biology, anywhere response timing matters. This could be in transcription/translation, post-translational signaling pathways, metabolism, vision, behavior, etc.

6.2 SREBP-1c degradation by the numbers

SREBP-1c undergoes phosphorylation that promotes recognition by an E3 ubiquitin ligase, which ubiquitinates SREBP-1c and promotes its proteasomal degradation. The phospho-mimetic mutant has a half-life of $t_{1/2} \approx 2$ hrs in cells while the WT sequence has a $t_{1/2} \approx 3$ hrs, as measured by a cycloheximide chase assay (Dong et al. Bioscience Rep 2016).

Since these measurements were conducted in slowly-dividing rat cells, $t_{1/2}$ values mostly reflect degradation. These values imply

$$\alpha_{\text{deg,WT}} \approx \ln(2)/3 \approx 0.23 \text{ hr}^{-1} \qquad \alpha_{\text{deg,mut}} \approx \ln(2)/2 \approx 0.35 \text{ hr}^{-1}$$

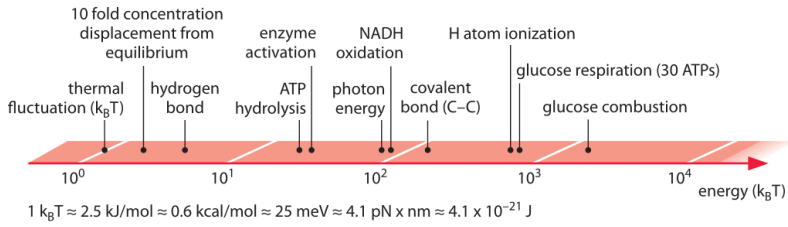

Figure 8: Energy scales in biology in units of k_BT.

For human cells, a typical transcription factor concentration is $10^4 - 10^5$ copies/cell.

$$\approx 0.3 \times 10^5 \approx f \times 10^4 \text{ copies degraded/hr/cell}$$

Compare this to the cellular capacity for protein degradation estimated in the Fig. 2 shows that the proteasome capacity exceeds SREBP-1c turnover flux by $\tilde{3}$–4 orders of magnitude.

$$10^5 - 10^6 \frac{\text{proteasomes}}{\text{cell}} \times 5 \frac{\text{proteins}}{\text{min. proteasome}} \times \frac{60\text{min}}{1 \text{ hr}} \approx f \times 10^7 - f \times 10^8 \text{ proteins/hr}$$

6.3 Degradation and Response Times

Degradation rates set the timescale on which protein or mRNA levels respond to induction. Degradation can be active (e.g., via the proteasome or RNAses), passive (spontaneous breakdown) or due to dilution by growth (making more cell volume reduces the concentration). The **effective rate of degradation** is governed by the faster of degradation and cell-cycle dilution.

	Degradation time of mRNA	Protein half-life (median)	Cell-cycle time
Bacteria	$\sim 3\,\text{min}$	~ 1–$2\,\text{h}$ ($\sim 1\,\text{h}$)	$\sim 1\,\text{h}$
Yeast	$\sim 30\,\text{min}$	$\sim 20h\,\text{min}$–$2\,\text{h}$ ($\sim 1\,\text{hr}$)	$\sim 3\,\text{h}$
Mammalian cell line	$\sim 300\,\text{min}$	$\sim 30\,\text{min}$–$2\,\text{days}$ (~ 9–$46\,\text{h}$)	$\sim 1\,\text{day}$

Table 1: Typical cellular timescales. mRNA response time $\approx \min(\tau_{\text{deg}}, \tau_{\text{cell cycle}})$. Protein lifetimes span from short-lived regulatory proteins to long-lived structural proteins; the wide mammalian range reflects both rapidly turned-over signaling proteins ($\sim$30 min) and stable abundant proteins ($\sim$days).

As shown in Table 1, mRNA lifetimes (τ_{deg}) are typically much shorter than the cell cycle time ($\tau_{\text{cell cycle}}$). Most mRNA degradation is catalyzed.

Most bacterial proteins, by contrast, are longer-lived and their measured half-lifes are determined by the cell cycle time (≈ 1 hr). In human cells, there is considerable variation among protein half-lives.

See a relevant vignette from the book: how fast to mRNAs and proteins degrade?

7 300 kJ/mol/bond – a lot or a little?

In the previous lecture we saw that metabolism breaks covalent bonds (extracts their electrons) and moves them to O_2. We calculated that a bond is 'worth' ≈ 300 kJ/mol. **Is this a lot of a little?**

What should we compare to? What is the right yardstick?

8 Thermal energy, Stokes-Einstein

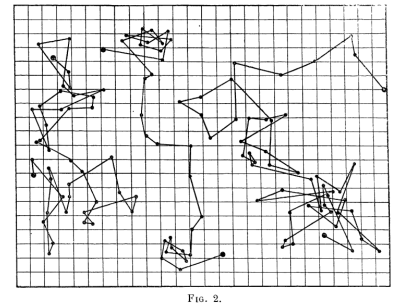

Figure 9: Colloidal particle traces by J. Perrin, 1909.

In lecture 4 we talked about Jean Perrin, Robert Brown and jiggling pollen grains.

Einstein had calculated D for a spherical particle

$$D = \frac{k_B T}{6\pi\eta r}$$

You can get $D \propto \frac{k_B T}{\eta r}$ by dimensional analysis (see lecture 4 notes) but you need to solve the Stokes drag problem to get the prefactor of $\frac{1}{6\pi}$. This is what Einstein did: used Stokes' expression for viscous drag on a spherical particle ($F = 6\pi\eta r v$) to derive a friction coefficient $\gamma = 6\pi\eta r$.

Notice the factor of $k_B T$ in the numerator – D increases with $k_B T$

- $T \approx 300$ K is the absolute temperature in Kelvin
- $k_B = 1.38 \times 10^{-23}$ J/K.
- $k_B T$ has energy units (J)
- $k_B T \approx 4 \times 10^{-21}$ J at 300 K

$k_B T$ is the **Thermal Energy Scale**.

D increases with T – the pollen grains will be more jiggly at higher T. The Boltzmann constant, k_B, translates temperature into energy units and tells us how much more jiggly.

8.1 $k_B T$ per mole

Multiply by N_A to get an energy scale per mole

$N_A k_B T \approx 6{\times}10^{23}$ molecules/mol$\times 4{\times}10^{-21}$ J/molecule ≈ 2500 J/mol $= 2.5$ kJ/mol

We see that 300 kJ/mol is a lot – it's more than $100\times$ the thermal energy scale.

Rule of Thumb: Thermal energy scale

$k_B T \approx 2.5$ kJ/mol

It's also useful to note the relationship between k_B and the gas constant

$$R = N_A \cdot k_B \approx 8.3 \text{ J/mol/K} = 8.3 \times 10^{-3} \text{ kJ/mol/K}$$

8.2 Estimating D for glucose

For glucose, the molecular radius is ≈ 0.5 nm. The challenge here is all in the units of viscosity. The viscosity of water is 1 centi-Poise which is

$$\eta = 1 \text{ cP} = 10^{-3} \mathrm{kg\,m^{-1}\,s^{-1}}$$

$$D \approx \frac{k_B T}{6\pi\eta r} = \frac{4 \times 10^{-21} \text{ J}}{20 \times 10^{-3} \mathrm{kg\,m^{-1}\,s^{-1}} \times 5 \times 10^{-10} \mathrm{m}} \approx \frac{4 \times 10^{-21} \mathrm{kg\,m^2\,s^{-2}}}{10^{-11} \mathrm{kg\,s^{-1}}} \approx 4{\times}10^{-10} \mathrm{m^2\,s^{-1}}$$

Converting to $\mu\mathrm{m^2\,s^{-1}}$

$$4 \times 10^{-10} \mathrm{m^2\,s^{-1}} \left(\frac{10^6 \mu\mathrm{m}}{1 \text{ m}}\right)^2 \approx 4 \times 10^2 \mu\mathrm{m^2\,s^{-1}}$$

not far from the book's value of $600\ \mu\mathrm{m^2\,s^{-1}}$.

Try doing the calculation for a protein.

9 Ligand binding

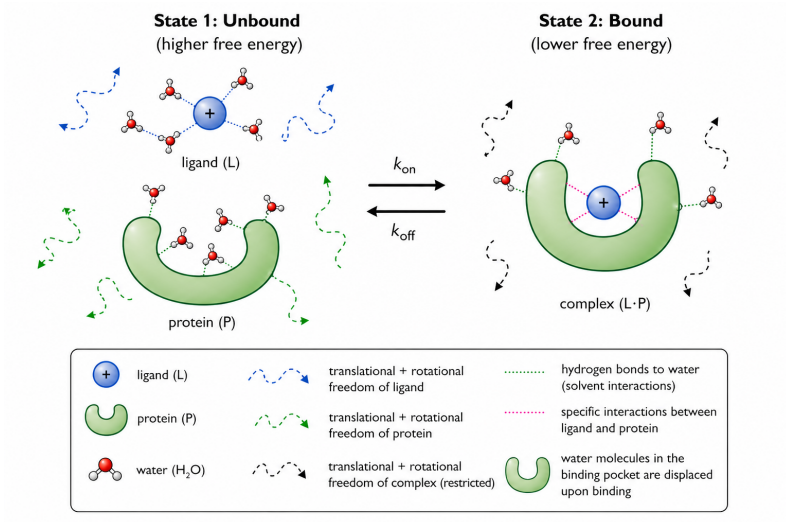

Figure 10: Favorable binding of a ligand by a protein. Since the molecules move together in the bound state, binding reduces the rotational and translational freedom of the two molecules. Assuming that ligand binding displaces all water interactions on both the ligand L and protein receptor P. This need not be the case in general, e.g., if some bridging waters are retained. Graphic made with ChatGPT

Think about a small molecule binding to a protein. The small molecule has a thermal energy $k_B T$ while the binding pocket has a bunch of water molecules bound to it. We often think of a dissociation constant

$$K_D = \frac{[L][P]}{[L \cdot P]} \text{ (molar units)} \qquad \Delta G_{\text{bind}} = RT \ln(\frac{K_D}{1\text{M}}) \qquad K_D = \exp\left(\frac{\Delta G_{\text{bind}} \cdot 1\text{M}}{RT}\right)$$

Typical protein-ligand interactions have $K_D \approx 10^{-9} - 10^{-7}$ molar.

$$\Delta G_{\text{bind}} = RT \ln(10^{-9}) \approx -20 \times 2.5 \approx -50 \text{ kJ/mol}$$

Reminder: the negative sign connotes a favorable binding interaction – bound state is more stable than unbound.

9.1 Fraction bound

You might remember that you can relate the fraction of protein that is bound to the K_D as follows

$$f_{\text{bound}} = \frac{[L \cdot P]}{[P] + [L \cdot P]} = \frac{[L]}{[L] + K_D}$$

This is an equilibrium relationship. Try to derive it. It's can be framed in terms of ΔG_{bind}.

$$\frac{[L \cdot P]}{[P] + [L \cdot P]} = \frac{[L]}{[L] + \exp\left(\Delta G_{\text{bind}}/RT\right)}$$

9.2 Energy content in concentration changes

Before we move on let's notice that we could have written

$$\Delta G_{\text{bind}} = -9 \times (RT \ln(10)) \approx -9 \times 5.8 \approx -50 \text{ kJ/mol}$$

Notice that each order of magnitude change in K_D alters ΔG by $RT\ln(10) \approx 6$ kJ/mol. From this we get a useful rule of thumb

Rule of Thumb: OOM Energy content

An order of magnitude separation between products and substrates is "worth" ≈ 6 kJ/mol

This rule applies in either direction. If binding or reaction becomes more less favorable, we add 6 kJ/mol (more positive ΔG). If binding/reaction becomes more favorable, we add -6 kJ/mol. **More negative ΔG values are more favorable** by convention.

9.3 Estimating the binding free energy

Can we understand where the ≈ -50 kJ/mol comes from?

As shown in Fig. 10, binding restricts the freedom of the protein and ligand. It also replaces interactions with solvent (water) by interactions between the ligand and protein. The free energy change associated with binding therefore reflects a balance:

$$\Delta G_{\text{bind}} = (\text{new ligand–protein interactions}) - (\text{interactions with water}) - (\text{loss of freedom})$$

Each of these contributions is typically on the order of a few k_BT (≈ 10 kJ/mol). Reaching a total of ~ 50 kJ/mol requires many such contributions adding up.

For example: - hydrogen bonds: ~ 5–10 kJ/mol each - electrostatic interactions: similar scale - entropy loss: costs a few k_BT

$$\underbrace{-50 \text{ kJ/mol}}_{\Delta G_{\text{bind}}} = (\text{new ligand–protein interactions}) - \underbrace{f \times \Delta G_{\text{H-bond}}}_{\text{lost water interactions}} - \underbrace{(-f \times k_BT)}_{\text{lost freedom of motion}}$$

So strong binding (nanomolar K_D) therefore the cumulative effect of multiple interactions, each comparable to the thermal energy scale.

To estimate the number and/or strengths of these interactions, we need a reference energy for the ligand-water and protein-water hydrogen bonds displaced upon binding.

10 Energy in an H-bond

Biology is full of transient interactions at the molecular scale. **Hydrogen bonding is a very important form of transient (non-covalent) interaction in biology** – water with biomolecules, water with water, biomolecules with each other, etc.

When H is bonded w/ a very electronegative atom like Oxygen, electron density moves towards that atom, creating a "dipole". These bonds can form spontaneously between water molecules. They are actually also important for:

- Protein secondary structures (α-helix, β-sheets).
- Transcription factor binding.
- DNA base pairing.
- RNA folding.
- protein aggregation.
- the hydrophobic effect.

For example, **H-bonds hold double stranded DNA** – there are more H-bond partners for complementary base-pairs (G-C, A-T) than non-complementary ones.

As discussed in the book, the energy content of a hydrogen bond is 2-12 k_BT.

Rule of Thumb: H bond energy

An H bond energy is $\approx f\, k_BT \approx 10$ kJ/mol

Returning to the protein-ligand interaction

$$\text{(new ligand–protein interactions)} \approx \underbrace{-50 \text{ kJ/mol}}_{\Delta G_{\text{bind}}} + \underbrace{f \times \Delta G_{\text{H-bond}}}_{\text{lost water interactions}} + \underbrace{(-f \times k_BT)}_{\text{lost freedom of motion}}$$

$$\text{(new ligand–protein interactions)} \approx -50 \text{ kJ/mol} - f \times 10 \text{ kJ/mol} - f \times 2.5 \text{kJ/mol} \approx -(80-100) \text{ kJ/mol}$$

So a few "typical" H-bonds is not enough. The binding interaction needs to be formed of stronger interactions or more of them.

11 Ribosome specificity – equilibrium model

The ribosome translates RNA to protein using RNA-RNA interactions that are due to H-bonds.

Lets compare a complementary G=C base pair, which "match," and a non-complementary G=U pair. These differ in 1 H bond, but could be read out as different amino acids.

- State 1: G=C.
- State 2: G=U (Mismatch).
- $\Delta\Delta G \approx f \times k_BT \approx 10$ kJ/mol

There is one missing hydrogen bond. What is the ΔE between the states?

Applying our rule of thumb, we see that the complementary G=C pair should be favored over G=U by

$$K = \exp(10/RT) \approx \exp(4) = 50 \text{ fold}$$

Using $2k_BT$ would give tenfold, while a higher number might give 100 fold. Yet measurements of ribosome accuracy give misincorporation rates of $\approx 10^{-3} - 10^{-4}$ (Kramer 2007, Mordret et al., 2016)

This observation motivated classic papers asking how the ribosome might use energy to reduce its error rate below the value implied by the equilibrium thinking diagrammed above (Hopfield 1974, Ninio 1975).

We could make a similar argument regarding DNA basepairing, where the error rate is even lower (1 error in $10^9 - 10^{10}$).

12 The Hydrophobic Effect

We talking about equilibrium binding above, where to ratio of two states occupancies (probabilities) was related to a free energy difference

$$\frac{P_1}{P_2} = e^{-\Delta G/RT}$$

- $\Delta G = G_1 - G_2$: free energy difference
- $RT = N_A k_B T$: thermal energy scale ≈ 2.5 kJ/mol

If $\Delta G > 0$ and $\frac{P_1}{P_2}$ small $\implies$ P_2 much preferred over P_1. In other words, lower energy states are more likely than higher energy states in thermodynamic equilibrium.

We also talked about hydrogen bonds, e.g. $\sim 2 - 12 k_B T$ in different contexts. Weak bonds that form between dipole molecules. E.g. DNA base pairing.

What happens if we throw a polar molecule into the mix?

- By itself, a favorable (low energy) configuration of H_2O molecules is to form as many H-bonds as possible w/ its neighboring H_2Os.
- A non-polar molecule disrupts that. H-bonds cannot be formed w/ non-polar molecules.
- This has the effect of decreasing the number of ways H_2O can arrange itself.

There are many ways for H_2O to orient themselves and still get H-bonds w/ neighbors. Introducing a non-polar molecule to the mix means LESS ways to configure the water. Water needs to arrange itself around the non-polar molecule. Water must become ordered. This is the essence of the **Hydrophobic Effect**.

When there are fewer ways for the system to be configured, it is less likely and, therefore, a higher energy state. The system tries to find ways to reduce the energy associated with ordering water.

Energy can be reduced by mashing hydrophobic molecules together ("burying" them).

Rule of Thumb: Hydrophobic effect

Hydrophobic energy penalty is about $4k_BT/\text{nm}^2 \approx 10$ kJ/mol/nm^2.

The energy penalty scales w/ exposed surface area because H_2Os along the molecule's interface are affected.

12.1 Example of binding a methyl group in a hydrophobic pocket.

- State 1: Unbound. Exposed SA $\approx 1\text{nm}^2$.
- State 2: Bound. "Burying" the hydrophobic group much lower energy, preferred.

A 1 nm^2 area $\implies \Delta G_{\text{bind}} \approx -10$kJ/mol. The bound state is favored by ≈ 50 fold at equilibrium.

12.2 Example of O_2 in H_2O

Dioxygen, or O_2 is non-polar with a radius ≈ 0.2 nm.

$$SA \approx 6 \times R^2 = 6 \times (0.2 \text{ nm})^2 \approx= 0.24 \text{ nm}^2$$

The energy penalty of O_2 floating around vs. burying it (e.g. w/ other O_2 molecules) is:

$$10 \text{ kJ/mol/nm}^2 \times 0.24 \approx 2.4 \text{ kJ/mol} \approx RT$$

About the same as the thermal energy scale. For O_2, the hydrophobic effect is about as strong as thermal fluctuations, meaning no huge penalty. Indeed O_2 is reasonably soluble in water with $\approx 200 - 300 \mu\mathrm{mol}\,\mathrm{dm}^{-3}$ concentrations at equilibrium with atmosphere (21% O_2) around 300 K.

13 The energy content of ATP hydrolysis

ATP formation/hydrolysis is the first thing that comes to mind when discussing biological energy.

$$\text{ATP} + H_2O \leftrightarrow \text{ADP} + P_i \quad \text{(ATP Hydrolysis)}$$

Because the phosphate is a "good leaving group" the hydrolysis reaction is "downhill" (products have lower free energy) and such reactions can be coupled to other biological processes to do work.

How much energy is there in this hydrolysis reaction? How does that compare to the other values we discussed, e.g., the thermal energy scale and the 300 kJ/mol bonds?

You might remember this expression connecting the reaction equilibrium constant K_{eq} to the standard-state reaction free energy ΔG°

$$K_{eq} = \frac{[\text{ADP}]_{eq}[P_i]_{eq}}{[\text{ATP}]_{eq}} \qquad K_{eq} = e^{-\Delta G^\circ/RT}$$

- K_{eq}: equilibrium constant.
- R: Gas constant
- ΔG°: Standard Gibbs Free Energy.

Notice the similarity with the K_D expression above and also the sign difference. The reason for the difference is that K_D is defined for the dissociation reaction, so K_D has a small value when binding is favorable (the bound state minimizes the free energy). K_eq has a large value when the reaction is favorable (i.e., when the products are the abundant species at equilibrium).

Note also that we can neglect H_2O as a reactant in determining K_{eq}. This is like setting its concentration $[\mathrm{H_2O}] = 1$. This is a convention in biochemical thermodynamics. We can do this because the water concentration barely changes between conditions, and so it will not affect the value of ΔG or K_{eq}.

In general, ΔG and K_{eq} this depends on the conditions (pH, ionic conc, etc.). The convention is to look at equilibrium in a biochemical standard state with pH 7.0, $25°$C and physiological ion concentrations. The biochemical standard state is denoted $\Delta G^{\circ\prime}$ (vs ΔG° "Chemical Standard State"). You can read more in Alberty et al. 2011.

Solving for the standard free energy gives:

$$\Delta G^{\circ\prime} = -RT \ln(K_{eq})$$

- $R \approx 8$ J/mol K
- $T \approx 300$ K
- $RT \approx 2.5$ kJ/mol

ATP Hydrolysis has

$$\Delta G^{\circ\prime} \approx -30 \text{ kJ/mol}$$

Applying our rule of thumb, we can translate -30 kJ/mol $= -5 \times 6$ kJ/mol to $K_{eq} \approx 10^5$. So at equilibrium there will be much more ADP than ATP.

13.1 ATP in bacterial cells

When we look at ATP and ADP concentrations inside cells, we find that **cells are far from equilibrium**. In cells, [ATP] is much higher [ADP]. In *E. coli*:

- ≈ 10 mM ATP
- 0.6 mM ADP
- 20 mM P_i

Away from equilibrium, the reaction quotient

$$Q = \frac{[\text{ADP}][P_i]}{[\text{ATP}]} \neq K_{eq}.$$

These quotients are only equal in equilibrium. Based on *E. coli* measurements,

$$Q = \frac{6 \times 10^{-3} \times 2 \times 10^{-2}}{10^{-1}} \approx 10^{-3} \text{ M}$$

What this says is cells are FAR from equilibrium when it comes to ATP. This is by design...

The apparent free energy of ATP hydrolysis is calculated as

$$\Delta G' = \Delta G^{\circ\prime} + RT \ln(Q)$$

You can see how this relates to the equilibrium formula we had before. If $Q = K_{eq}$, then $\Delta G' = 0$, meaning equilibrium. No energy to be gained from ATP in equilibrium.

Let's apply our rule of thumb (6 kJ/mol per order of magnitude) to estimate the $\Delta G'$ of ATP hydrolysis in *E. coli*.

$$\Delta G' = -30 \frac{\text{kJ}}{\text{mol}} - 3 \times 6 \text{ kJ/mol} \approx -50 \text{ kJ/mol}$$

Since $\Delta G' < 0$ the hydrolysis reaction is favorable and can be coupled to chemical or mechanical work.

13.2 ATP in human cells

The $\Delta G' < 0$ of ATP hydrolysis depends on the cell because reactant concentrations will depend on the cell and conditions. In human muscle cells:

- ATP ≈ 8 mM
- ADP $\approx 9\mu$M
- $P_i = 4$ mM

$$Q = \frac{9 \times 10^{-6} \times 4 \times 10^{-3}}{8 \times 10^{-3}} \approx f \times 10^{-6}$$

$$\Delta G = -30 - 2.5 \ln(f \times 10^{-6}) \text{ kJ/mol} \approx -60 \text{ kJ/mol}$$

So ATP "buys more" in the muscle.

13.3 Generality of reaction thermodynamics

$\Delta G = \Delta G^{\circ\prime} + RT \ln(Q')$ is not just about ATP. It can be applied to any reaction.

You can find $\Delta G^{\circ\prime}$ values for common reactions easily in eQuilibrator and other resources.

For example, consider the malate dehydrogenase reaction in the TCA cycle:

$$\text{Malate} + \text{NAD}^+ \rightarrow \text{Oxaloacetate} + \text{NADH} + \text{H}^+.$$

$$\Delta G^{\circ\prime} \approx +30 \text{ kJ/mol}$$

Presumably this unfavorable reaction in the heart of central metabolism has some consequences for cells. What are some ways that cells might overcome this thermodynamic barrier?